TECHNICAL WHITE PAPER

AI-Optimized Continuity Test Time Reduction for Semiconductor Devices

Open/short (contact) test optimization on automatic test equipment using conventional test-program techniques combined with machine-learning models for settling-time prediction, risk-based pin ordering, adaptive flows and intelligent retest control.

Document	Technical white paper / implementation guide
Scope	Wafer sort (CP) and final test (FT) continuity suites
Applicable ATE	Platform-independent (PPMU-equipped digital testers)
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1. Executive Summary

The continuity test (also called the open/short or contact test) is the first test executed on every semiconductor device at wafer sort and at final test. Its purpose is to prove that every device pin is electrically connected to the tester through the probe card or socket, and that no pin is shorted to a supply or to a neighbour, before any power is applied. Because it runs on 100% of devices, at every insertion and on every pin, it is a recurring cost that scales directly with pin count and volume.

On high-pin-count devices tested with a conventional serial approach (one shared parametric measurement unit stepping through pins one at a time with fixed worst-case waits) the continuity suite can consume from several hundred milliseconds to over a second per site. This paper describes a layered strategy to reduce that time:

- **Layer 1 – Test-program engineering:** parallel per-pin measurement (PPMU), ganged group testing, relay and range discipline, and multisite concurrency. These deliver the largest single reduction.
- **Layer 2 – AI/ML optimization:** learned per-pin settling times, failure-probability-ordered (fail-fast) testing, adaptive flows that skip redundant measurements on stable material, classifiers that separate contact problems from real device defects, and outlier detection that improves quality at no extra test time.
- **Layer 3 – Governance:** guard bands, correlation studies, fallback logic and customer/quality approval so that time saved never becomes escape risk.

A modelled 512-pin device used throughout this paper goes from **1233 ms** to **19.4 ms** per site (a **98.4%** reduction) when the layers are combined. The figures are derived from a parametric time model and simulated data; real savings depend on tester architecture, pin count, contact quality and product maturity, and should be confirmed with a baseline study on the target platform (Section 9).

> **Key takeaway**
> Parallelization removes most of the time; AI then removes the conservative margins that parallelization leaves behind (worst-case waits, redundant polarities, blind retests) while adding quality insight. Neither should reduce fault coverage without an explicit, validated and approved risk decision.

2. Continuity Test Fundamentals

2.1 Purpose

Before a device is powered, the tester must know that each channel is actually touching its pin. A missing contact would make every later test result meaningless, and powering a device with a shorted pin can damage the device, the probe card or the tester. The continuity test therefore answers three questions for every signal pin: is it connected (no open), is it isolated (no short to supply or ground), and is it isolated from adjacent pins (no pin-to-pin short).

2.2 Measurement principle

Almost every CMOS I/O pad carries electrostatic-discharge (ESD) protection: one diode from the pad to VDD and one from VSS to the pad. With all supplies held at 0 V, the tester forces a small current into or out of the pin under test and measures the resulting voltage. Pulling roughly 100 µA out of the pin forward-biases the VSS-side diode, so a properly contacted pin shows a negative diode drop of about 0.5–0.7 V. Pushing current in forward-biases the VDD-side diode and gives a positive drop.

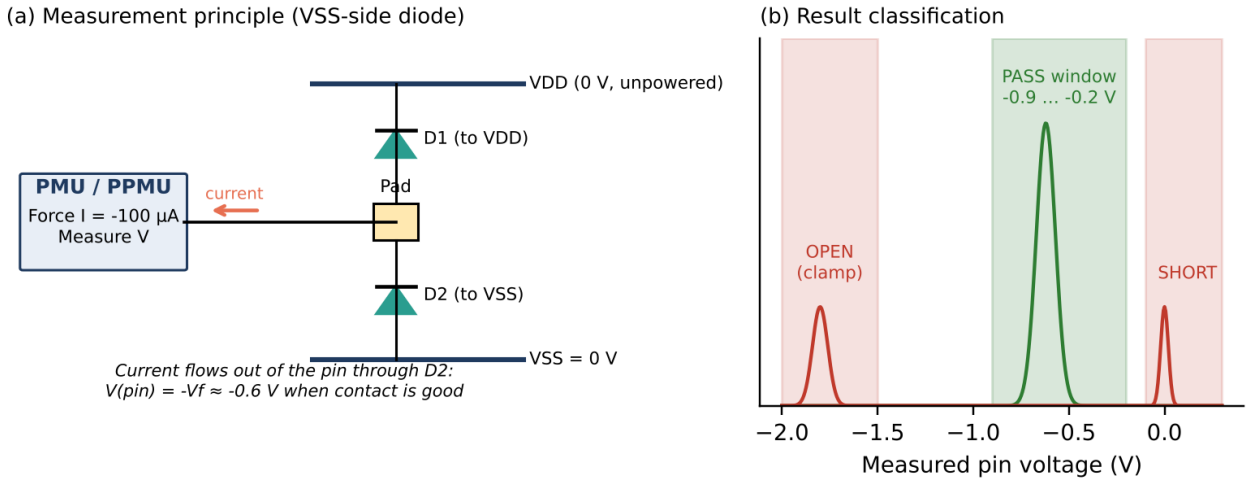


Figure 1 – Continuity measurement through the ESD protection diodes (a) and classification of the measured voltage into open, pass and short regions (b).

2.3 Typical test conditions and limits

Parameter	Typical value	Notes
Forced current	−100 µA (VSS diode), +100 µA (VDD diode)	50–500 µA depending on pad design and leakage
Voltage clamp	±1.5 to ±2.0 V	Protects the pad if the pin is open; open reads at the clamp
Pass window (VSS side)	−0.90 V ≤ V ≤ −0.20 V	Set from characterization data, not datasheet nominal
Open criterion	|V| ≥ ~1.4 V	No current path: voltage rises to the clamp
Short criterion	|V| ≤ ~0.10 V	Pin tied to ground, a supply or an adjacent driven-low pin
Other pins during test	Held at 0 V by their drivers or PPMU	Required so that pin-to-pin shorts become visible
Supplies	VDD/VSS at 0 V, supplies connected	Device unpowered

2.4 Variants

- **Single-polarity test:** only the VSS-side diode is checked. Fastest, detects opens and most shorts.
- **Dual-polarity test:** both diodes are checked. Detects a damaged or missing VDD-side diode and some supply shorts.
- **Pin-to-pin short test:** adjacent pins are driven to a different level while the pin under test is measured, or a matrix pattern is used. Often combined into the main test by holding all other pins at 0 V.
- **Power-pin continuity:** checks supply pins through the device power-supply channels; usually a separate, short test.
- **Kelvin (force/sense) continuity:** for probe cards or sockets with Kelvin contacts, verifies both the force and sense paths, typically by resistance measurement.

3. Anatomy of Continuity Test Time

3.1 Time model

For a serial implementation the continuity time per site can be expressed as:

```
T_cont = T_setup + N_pins × N_pol × ( t_relay + t_range + t_settle + t_meas ) + T_retest

  N_pins   number of signal pins tested
  N_pol    number of polarities (1 or 2)
  t_relay  relay switching + debounce per pin (connect PMU, disconnect)
  t_range  PMU range change, if any
  t_settle wait for the forced node to reach a stable voltage
  t_meas   ADC conversion / averaging
  T_retest additional time from retest of failing or marginal units (averaged over all units)
```

For a parallel PPMU implementation with G measurement groups the product term becomes G × N_pol × (t_connect + t_settle,group + t_meas), where t_settle,group is set by the slowest pin in the group. The key observation is that the waits (relay debounce and settling) dominate, not the measurement itself.

3.2 Baseline for the reference device

Element	Per pin, per polarity	Total (512 pins × 2)	Share
Relay switching + debounce	0.30 ms	307.2 ms	24.9%
Settling (fixed worst case)	0.80 ms	819.2 ms	66.5%
Measurement / ADC	0.10 ms	102.4 ms	8.3%
Setup and teardown	—	4.0 ms	0.3%
Total	**1.20 ms**	**1232.8 ms**	**100%**

Two thirds of the baseline is waiting for the node to settle with a delay chosen to cover the slowest pin under the worst conditions. That conservative margin is exactly what machine learning can recover once parallelization has removed the per-pin multiplication.

3.3 Hidden contributors

- **Retests:** a contact failure followed by an automatic retest doubles that unit's continuity time and adds handler or prober index time. At 2–5% retest rates this is significant and is often invisible in per-test time reports.
- **Range switching:** mixing current or voltage ranges between pins forces internal instrument relay changes.
- **Datalogging:** logging every pin result to the datalog in production can add measurable overhead on some platforms.
- **Site serialization:** if the continuity suite uses a shared resource, sites run one after another instead of concurrently.
- **Probe-cleaning and contact recovery:** stops triggered too late or too often both cost throughput.

4. Conventional Test-Time Reduction Techniques

These techniques should be applied first; they are deterministic, well understood by quality teams and deliver most of the saving. The AI methods in Section 5 build on top of them.

4.1 Parallel per-pin PMU measurement

Modern digital testers provide a parametric measurement unit behind every digital channel (PPMU). Instead of routing one shared PMU to each pin in turn, all pins in a group are forced and measured simultaneously. With 64 pins per step the 1,024 serial measurements of the reference device collapse to 16 parallel steps. Limits are checked per pin, so diagnostic resolution is unchanged.

- Verify that PPMU accuracy and clamp behaviour meet the continuity limits (they usually do, since the window is wide).
- Check that simultaneous forcing on adjacent pins does not mask pin-to-pin shorts. If it does, split pins into interleaved groups (odd/even) so that neighbours are never forced together.
- Keep the number of simultaneous forcing channels within the instrument's power and thermal budget.

4.2 Ganged (group) testing with fail-only bisection

When per-pin parallel measurement is unavailable, pins can be tied together through the tester's ganging capability and measured as a group. A good group shows the expected combined response; only a failing group is subdivided to find the faulty pin. Because most units pass, average time is close to the number of groups, not the number of pins.

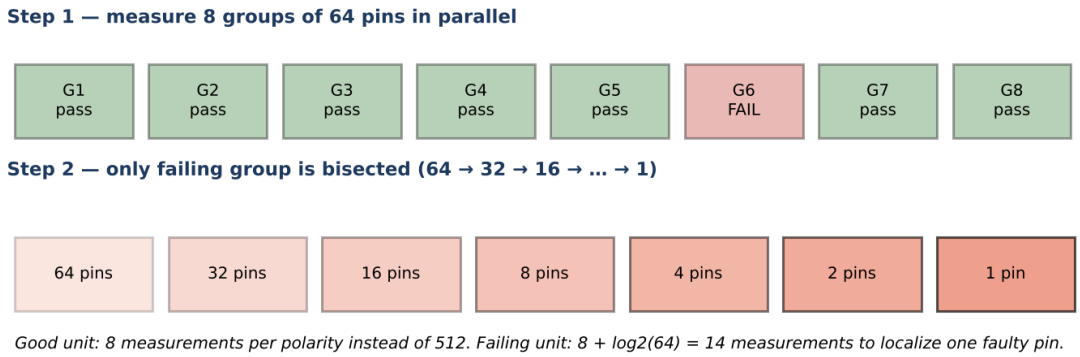


Figure 2 – Ganged test with bisection only on failure.

4.3 Digital-channel (functional) continuity

The pin electronics' active load or driver can source a current while the window comparator checks the resulting voltage. A short pattern then checks all pins at full digital speed. This is very fast and adequate for open detection; it is less precise for parametric screening and is often used as a fast first pass, with PPMU measurement only on failure.

4.4 Relay, range and sequence discipline

- Use one current range and one voltage clamp for the whole suite to avoid range changes.
- Connect all needed relays once at the start and disconnect once at the end; avoid per-pin connect/disconnect.
- Order pins to follow physical instrument boards, so that relay banks switch together.

- Replace fixed software waits with the instrument's hardware settling or ready flag where available.
- Disable per-pin datalogging in production; log only failures and summary statistics, with full logging on sampled units.

4.5 Multisite and concurrency

Continuity must run concurrently across sites. Shared resources (for example a single high-accuracy PMU) should be removed from the suite. Where the handler or prober index time is long, the continuity of the next touchdown can be prepared during indexing (pre-loading patterns and levels).

Technique	Typical saving	Fault coverage impact	Effort
Parallel PPMU	80–97% of continuity time	None (per-pin limits kept)	Medium
Ganged + bisection	70–90%	None if bisection is complete	Medium
Digital-channel continuity	85–98%	Reduced parametric precision	Low–Medium
Relay/range discipline	10–25%	None	Low
Multisite concurrency	Per-unit time ÷ sites	None	Medium–High

5. AI/ML Optimization Methods

After parallelization, the remaining time is mainly conservative waiting and redundant work done identically for every unit. Machine learning uses the large amount of data the tester already produces to make those decisions per pin, per lot and per touchdown. Five methods are described below, ordered from lowest to highest risk.

5.1 Learned settling time

Each pin presents a different RC load to the PPMU: trace length on the probe card or load board, pad capacitance, package routing and ESD structure all differ. A fixed wait must cover the slowest pin. A regression model trained on characterization data predicts the settling time each pin (or group) actually needs, and the program uses the prediction plus a guard band.

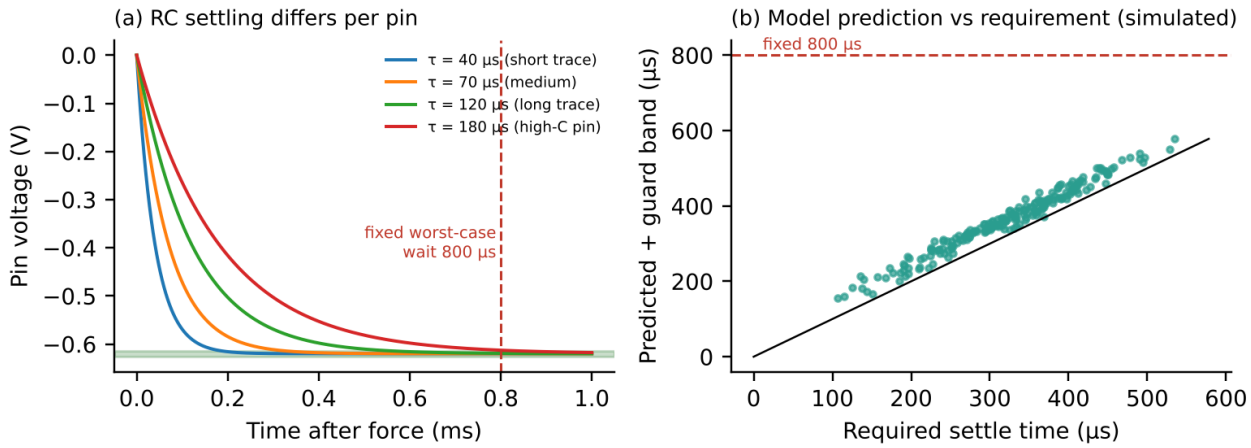


Figure 3 – (a) Settling curves for pins with different time constants. (b) Simulated model predictions with guard band against required settling time.

Data collection

- Run the continuity suite on a characterization sample (typically several hundred to a few thousand units across lots, sites and temperature) with a sweep of wait times, or capture the

voltage waveform with the PPMU's digitizer.

- For each pin and condition, find the shortest wait after which the reading is within a tolerance (e.g. 1% or 5 mV) of the fully settled value.
- Record features: pin name, instrument channel, load-board trace length, pin type, package, site, temperature, probe-card ID and touchdown count.

Model

Gradient-boosted trees or a simple linear model are sufficient; the relationship is physical and smooth. The production wait for a group is the maximum predicted value among its pins plus a guard band set from the model residuals (for example the 99.9th percentile of under-prediction, or three standard deviations, whichever is larger).

In the simulated study, a linear model on four features gave a guard band of 44 µs and an average guarded wait of 364 µs per pin against a fixed 800 µs, with 0 of 200 held-out pins under-predicted. At group level (slowest pin plus guard) this gives the 0.45 ms used in the time model.

Deployment

The model is not run on the tester. Its output is compiled into a static lookup table (wait per group, per site) that is reviewed, version-controlled and loaded by the test program. The table is regenerated when hardware changes (new probe card or load board revision) and checked periodically against sampled full-wait data.

5.2 Risk-ordered (fail-fast) pin testing

Contact failures are not uniform across pins. Corner and edge pins, pins at the end of long probe needles, worn socket pogo pins and specific site positions fail far more often than others. A ranking model estimates the probability that each pin will show an open or short on the next touchdown, and the test sequence is reordered so that high-risk pins are checked first. With stop-on-first-fail enabled, defective contacts are detected earlier.

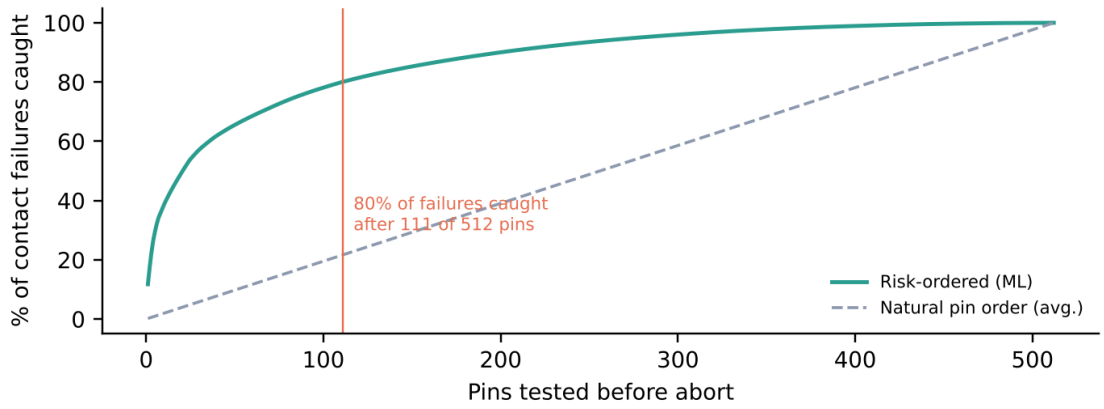


Figure 4 – Simulated cumulative capture of contact failures for risk-ordered versus natural pin order.

In the simulation (heavy-tailed failure distribution over 512 pins), the expected number of pins measured before a failing pin is found drops from about 257 with natural ordering to about 64 with risk ordering; 80% of failures are caught within the first 111 pins. The benefit applies only to failing units, so the average saving is modest when failure rates are low. Its main value is in serial or partially serial suites, in early detection of contact degradation, and in combination with Section 5.4.

Features

- Historical open/short counts per pin, per site and per probe-card or socket serial number
- Touchdowns since last clean and since last needle replacement or socket service

- Recent marginal readings (diode voltage drifting toward the open limit)
- Wafer position (edge die), lot, temperature and handler/prober ID

A logistic regression or gradient-boosted classifier, retrained daily or per shift, is adequate. The output is a pin order per site that changes slowly, loaded by the test program at lot start.

5.3 Adaptive flows

On a mature product with stable contact quality, parts of the continuity suite give almost no information on most units. Adaptive test uses real-time and historical data to decide when a reduced flow is safe and when the full flow must run.

Adaptive decision	Trigger for reduced flow	Trigger to return to full flow
Skip VDD-side polarity	VSS-side result clean, lot yield and contact history stable, model risk below threshold	Any VSS-side fail or marginal value; new lot, new probe card, after clean; periodic sample
Reduce pin-to-pin short check	No pin-to-pin shorts in N consecutive lots on this design	New mask set, new package, any short detected
Skip continuity at FT (done at CP)	Rarely acceptable; consider only with strong package-level data	Required for most products and all automotive

For the reference device, skipping the VDD-side polarity on 80% of units reduces the parallel suite from 29.6 ms to 19.4 ms. Every reduced flow must keep a sampling rate of full-flow units so that the model's assumptions are continuously verified, and the decision logic must default to the full flow on any data gap.

5.4 Contact-versus-device failure classification and retest control

When a pin fails continuity, the tester usually cannot tell whether the device is defective or the contact is poor. The common response, an unconditional retest, wastes time and can let a marginal contact pass by chance. A classifier uses the pattern of the failure to decide the best action.

Signal	Suggests contact problem	Suggests device defect
Spatial pattern	Same pin fails across consecutive dies or units	Random pin, isolated die
Site pattern	Concentrated on one site	Spread across sites
Voltage value	Near the clamp, unstable between readings	Repeatable short or specific abnormal value
History	High touchdown count since clean	Correlates with wafer region or lot
Recovery	Passes after re-touch or clean	Fails again after re-touch

Actions are then targeted: bin as a device fail without retest, retest once, trigger an early probe clean or socket inspection, or stop the lot for maintenance. Reducing unnecessary retests saves both test time and handler index time, and earlier cleaning prevents yield loss from clusters of false opens.

5.5 Outlier detection and dynamic part average testing

Continuity data contains parametric information. A diode voltage that passes the wide limits but sits far from the population of its wafer or lot may indicate a damaged ESD structure or a latent defect. Dynamic part average testing (PAT) and multivariate outlier models (for example robust Mahalanobis distance across all pins of a die) flag such units. This adds no measurement time because the data already exists; it converts the continuity suite from a pure gate into a quality screen, which helps justify the other optimizations to quality teams.

6. Combined Results on the Reference Device

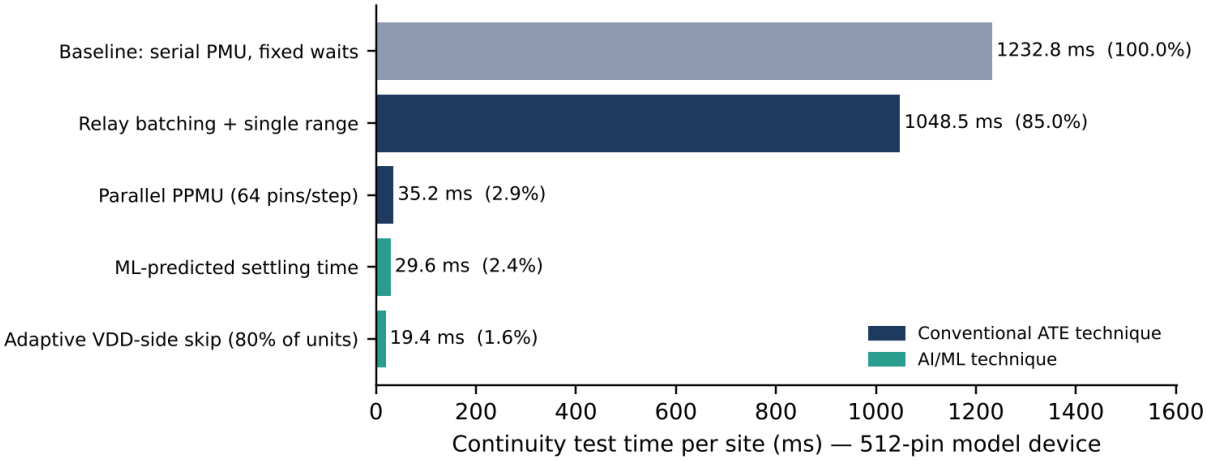


Figure 5 – Modelled continuity time per site after each optimization step (512-pin device).

Step	Change	Time (ms)	vs. baseline
Baseline: serial PMU, fixed waits	Serial PMU, 0.30 ms relay, 0.80 ms fixed settle, 0.10 ms measure, 2 polarities	1232.8	100.0%
Relay batching + single range	Relays connected once per bank, single range: 0.12 ms switching per pin	1048.5	85.0%
Parallel PPMU (64 pins/step)	64 pins per step, 1.0 ms connect, 0.15 ms measure	35.2	2.9%
ML-predicted settling time	Group wait from learned model + guard: 0.80 → 0.45 ms	29.6	2.4%
Adaptive VDD-side skip (80% of units)	VDD-side diode test skipped on 80% of units	19.4	1.6%

At 2 sites per wafer-sort touchdown and 30 million units per year, removing about 1.2 s of continuity time per unit saves roughly 5,000 tester hours per year at that insertion. Even the AI-only steps (35.2 → 19.4 ms) are worth about 66 tester hours a year per 30 million units, before counting reduced retests, which are usually the larger AI benefit in practice.

Interpreting these numbers
The reference device and parameters are illustrative. Use the time model in Section 3.1 with your own measured relay, settle and measure times, pin count, polarity count and retest rate to estimate the opportunity before starting a project.

7. System Architecture

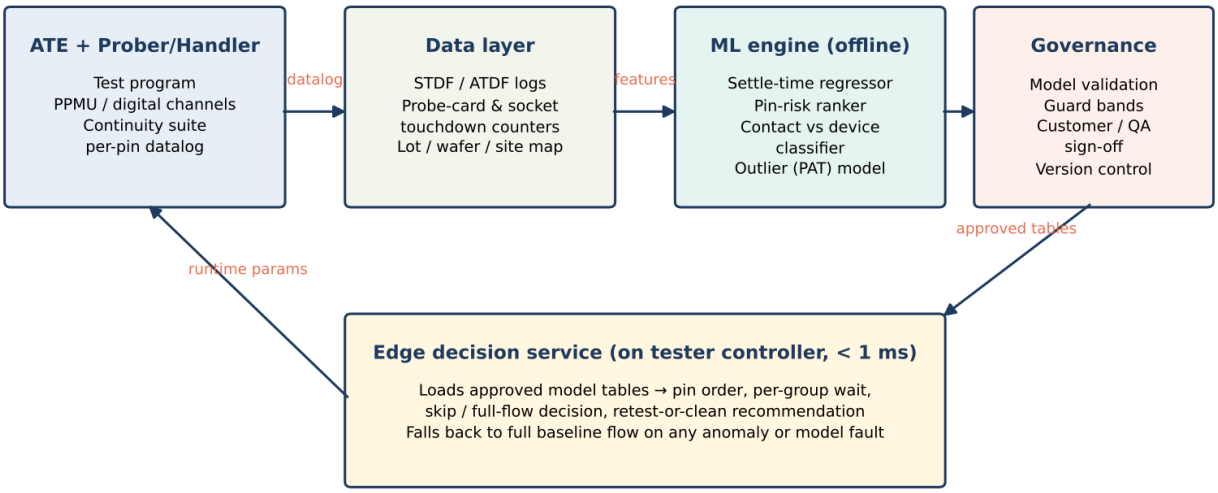


Figure 6 – Data and decision flow. Models are trained offline; only approved, static tables reach the tester.

7.1 Components

- **ATE and prober/handler:** the continuity suite writes per-pin values and pass/fail to the datalog, along with site, probe-card or socket ID and touchdown counters.
- **Data layer:** STDF files are parsed into a columnar store (for example Parquet) keyed by lot, wafer, die (x, y), site, pin and insertion. Maintenance events (clean, needle replacement, socket change) are joined in.
- **ML engine:** offline training of the settle-time regressor, pin-risk ranker, contact/device classifier and outlier model.
- **Governance:** validation reports, guard-band calculation, approval workflow and version control of every table released.
- **Edge decision service:** a small, deterministic component in the test program or on the tester controller that reads the approved tables and makes run-time choices in well under a millisecond. It falls back to the baseline full flow on any missing, stale or inconsistent input.

7.2 Design principles

- **Determinism on the tester:** no model inference with unbounded latency in the test flow; use lookup tables or tiny models.
- **Fail-safe defaults:** the full baseline flow is always available and is the default state.
- **Traceability:** every unit's datalog records the model/table version and the flow variant that was applied.
- **Separation of duties:** the ML team proposes; test engineering and quality approve and release.

8. Reference Algorithms

8.1 Ganged continuity with bisection

```
def continuity_ganged(pins, group_size, measure_group, limits):
    """measure_group(list_of_pins) -> True if all pins in list are within limits."""
    failing = []
    for g in chunks(pins, group_size):
        if measure_group(g):
            continue                        # whole group good: 1 measurement
        failing += bisect(g, measure_group)
    return failing

def bisect(group, measure_group):
    if len(group) == 1:
        return group                        # isolated failing pin
    a, b = group[:len(group)//2], group[len(group)//2:]
    bad = []
    if not measure_group(a): bad += bisect(a, measure_group)
    if not measure_group(b): bad += bisect(b, measure_group)
    return bad
```

8.2 Settling-time model and lookup-table generation

```
import pandas as pd, numpy as np
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.model_selection import GroupKFold

df = pd.read_parquet("settle_characterization.parquet")
# columns: pin, site, ppmu_group, trace_len_mm, pin_cap_pf, pin_type, temp_c, probe_card, t_required_us
X = pd.get_dummies(df[["trace_len_mm","pin_cap_pf","pin_type","temp_c","site"]])
y = df["t_required_us"]

model = GradientBoostingRegressor(n_estimators=300, max_depth=3, learning_rate=0.05)
resid = []
for tr, te in GroupKFold(5).split(X, y, groups=df["probe_card"]):   # hold out whole cards
    model.fit(X.iloc[tr], y.iloc[tr])
    resid.append(y.iloc[te] - model.predict(X.iloc[te]))
resid = pd.concat(resid)
guard = max(np.quantile(resid, 0.999), 3 * resid.std())

model.fit(X, y)
df["t_wait_us"] = model.predict(X) + guard
table = (df.groupby(["site", "ppmu_group"])["t_wait_us"].max()
           .clip(upper=800).round(-1))          # never exceed the original fixed wait
table.to_csv("continuity_wait_table_v1.csv")    # reviewed, versioned, loaded by test program
```

8.3 Risk-ordered pin sequence

```
from sklearn.linear_model import LogisticRegression

hist = load_contact_history(days=30)  # one row per (touchdown, site, pin)
feat = ["fail_rate_30d", "touchdowns_since_clean", "edge_die", "marginal_rate_7d", "needle_age"]
clf = LogisticRegression(max_iter=1000, class_weight="balanced").fit(hist[feat], hist["fail"])

current = current_state_per_site_pin()          # same features, as of lot start
current["p_fail"] = clf.predict_proba(current[feat])[:, 1]
order = (current.sort_values(["site", "p_fail"], ascending=[True, False])
                .groupby("site")["pin"].apply(list))
export_pin_order(order, version="risk_order_2026w42")
```

8.4 Adaptive VDD-polarity decision (runs on tester, table driven)

```
def run_vdd_polarity(unit, vss_results, ctx):
    if ctx.table_missing or ctx.table_stale:      return True   # fail-safe
    if any(r.fail or r.marginal for r in vss_results): return True
    if ctx.new_lot or ctx.after_clean or ctx.new_hardware: return True
    if ctx.unit_index % ctx.sample_every == 0:    return True   # audit sample
    return ctx.risk_score(unit) > ctx.threshold
```

9. Implementation Roadmap

Phase	Duration	Activities	Exit criteria
0. Baseline	2–3 weeks	Measure actual continuity and retest time per site from datalogs and test-time profiler; collect relay, settle and measure times; build the time model	Agreed baseline and opportunity estimate
1. Conventional	4–8 weeks	PPMU parallel or ganged suite; relay/range cleanup; datalog reduction; multisite review	Correlation: 100% pass/fail agreement with old suite on ≥ 10k units including known-fail samples
2. Data pipeline	4–6 weeks (parallel)	STDF parsing, maintenance event join, data quality checks, dashboards	Daily automated dataset
3. Settling model	4–6 weeks	Characterization sweep, model, guard band, wait table	Zero mismatches vs full wait on held-out units; GR&R; unchanged
4. Retest & ordering	6–8 weeks	Contact/device classifier, retest rules, risk ordering, cleaning triggers	Retest rate and false-open yield loss reduced without escape increase
5. Adaptive flows	8–12 weeks	Reduced-flow rules, sampling plan, customer approval	Approved deviation / PCN where required
6. Sustain	Ongoing	Drift monitoring, periodic re-validation, retraining on hardware change	KPIs stable

10. Validation, Risk and Quality Considerations

10.1 Correlation and repeatability

- Run old and new suites back to back on a large sample including devices with known opens and shorts (or deliberately lifted pins on a test socket) and require full pass/fail agreement.
- Repeat gauge repeatability and reproducibility (GR&R;) for the diode-voltage measurement with the shortened waits.
- Test across temperature, all sites, every probe card or socket in the fleet, and at both ends of the probe-card life.

10.2 Risks and mitigations

Risk	Mitigation
Shortened wait gives wrong reading on an untested hardware combination	Wait table keyed by hardware ID; unknown hardware uses fixed wait; periodic full-wait audit
Parallel forcing masks a pin-to-pin short	Interleaved groups; deliberate short injection during validation
Adaptive skip lets a VDD-diode defect escape	Sampling, rapid return to full flow, outlier screen; exclude automotive/high-reliability unless approved
Model drift (new probe card, process change)	Drift monitors on prediction residuals and failure rates; automatic fallback
Classifier mislabels a device defect as contact problem	Always rebin after retest result; never pass a unit without a clean full measurement
Traceability gaps	Log model and table version per unit; change control for every release

10.3 Customer and standards context

Any change that reduces the measurements performed on a shipped unit may require customer notification or approval under the applicable quality agreement. For automotive products, flows aligned with AEC-Q100 and the AEC-Q001 PAT guideline, and the requirements of IATF 16949, generally mean that parallelization and settle-time optimization are acceptable after correlation, while skipping measurements is much harder to justify. Confirm requirements with your quality organization.

11. Key Performance Indicators

KPI	Definition	Target direction
Continuity time per site	Mean suite time from test-time profiler	Down
Continuity share of total test time	Continuity time ÷ total test time	Down
Retest rate (contact)	Units retested due to continuity ÷ units tested	Down
False-open yield loss	Units failing continuity that pass on retest or after clean	Down
First-pass yield	Units passing all tests on first insertion	Up
Escape rate	Downstream or customer failures traceable to contact or pin defects	No increase (hard constraint)
Model health	Residual drift, under-prediction count, audit mismatches	Within control limits
Units per hour (UPH)	Throughput per tester	Up

12. Conclusion

Continuity testing is simple in principle but expensive in aggregate because it touches every pin of every unit at every insertion. The largest gains come from test-program engineering, above all parallel measurement, which typically removes the large majority of the time. AI then removes the conservative margins that remain, by tailoring waits to each pin, testing the most likely failures first, adapting the flow to material that has proven stable, and replacing blind retests with informed decisions. Implemented with guard bands, audit sampling, fail-safe defaults and quality approval, these methods reduce continuity time by well over 90% on high-pin-count devices while improving, rather than weakening, the insight the test provides.

Appendix A. Glossary

Term	Meaning
ATE	Automatic test equipment
CP / FT	Chip probe (wafer sort) / final test (packaged device)
PMU / PPMU	Parametric measurement unit / per-pin PMU behind each digital channel
ESD diode	Protection diode from pad to VDD or from VSS to pad; used as the continuity measurement element
Clamp	Voltage limit applied while forcing current, protecting an open pin
Ganging	Connecting several pins to one measurement resource at once
Settling time	Time for the forced node to reach a stable voltage after the current is applied
Guard band	Safety margin added to a model prediction or limit
Adaptive test	Changing the test flow per unit or lot based on data
PAT / DPAT	Part average testing / dynamic PAT: outlier screening relative to the population

Term	Meaning
STDF	Standard Test Data Format, the common ATE datalog format
GR&R;	Gauge repeatability and reproducibility
UPH	Units per hour
Touchdown	One contact of the probe card or socket with the device(s)

Appendix B. Data Fields Required

Source	Fields
Datalog (STDF)	Lot, wafer, die X/Y or unit ID, site, insertion, test number, pin, measured value, limits, pass/fail, test time, retest flag
Hardware	Tester ID, probe card / load board / socket serial, revision, PPMU channel mapping, trace length per pin
Maintenance	Clean events, needle replacement, socket service, touchdown counters
Characterization	Waveforms or wait-time sweeps per pin, temperature
Quality	Downstream fails, customer returns, failure analysis results linked to unit or lot